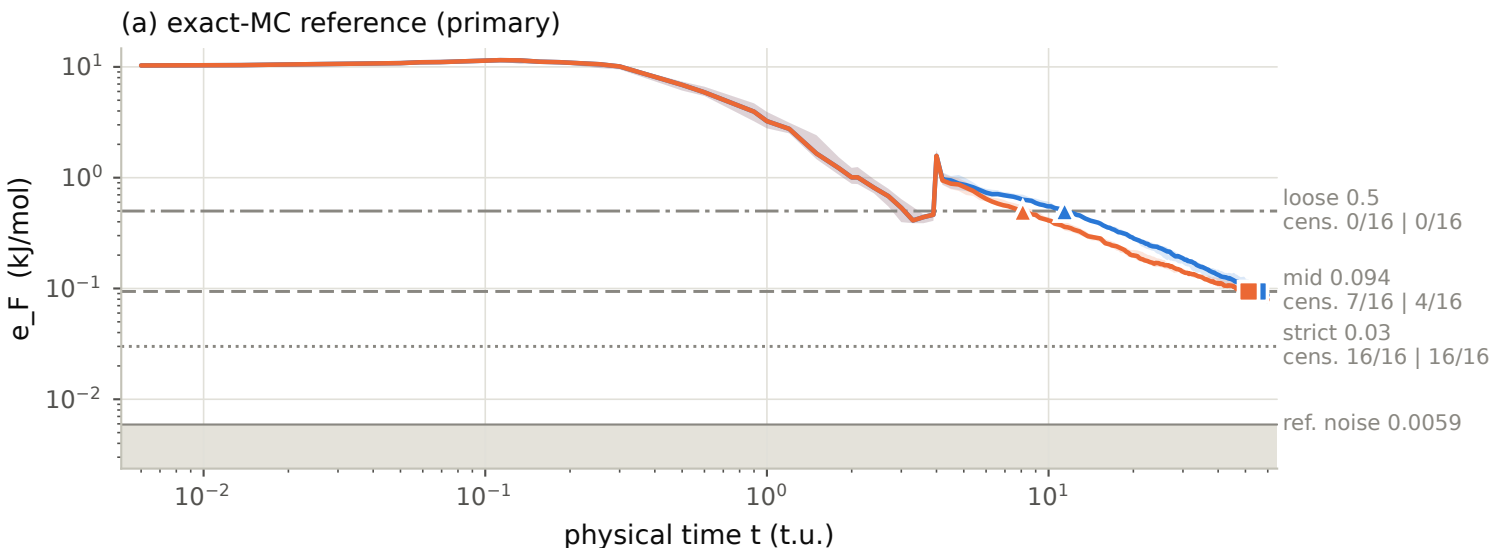


**Ethane/LTA 300 K · N = 1024 · T\_N = 60 t.u.**

\_steps = 300,000 · h = 0.0002 · B = N n\_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16, ABF+FR 16/16

Free-energy error  $e_F$  vs physical time



ABF (n = 16)      interquartile range      —□— mid threshold      MC reference noise (RMS of the per-k

ABF+FR (n = 16)      strict threshold      —△— loose threshold

Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC\_PLAN section-4 quantities, unsmoothed.

Markers: first save after which the seed-MEDIAN curve stays at or below the threshold at every later save (persistent crossing of the median curve; the per-seed tau statistics are in summary.json). Censored thresholds have no marker; 'cens. a | b' next to each threshold = seeds whose own tau is censored (never persistently met by  $u = 1$ ), ABF | ABF+FR.

Errors over the full circle (180 bins); Gamma =  $M / \max(C, 1)$  from the production accumulators once they hold counts (else all-steps), as core\_lta reports.

eqb\_plot\_config/2 · config lta\_300K\_production.json · 2026-10-10 07:06 UTC